

Agriculture: Apple Pest Management Guidelines

Fungicide Efficacy—Conventional Products

Fungicide***	Resistance risk (FRAC group number) ¹	Scab		Powdery mildew
		protectant	eradicant	
Adament**	medium (3/11) ³	5	5	5
Aprovia	high (7)	5	3	4
Axios	high (52)	4	4	5
Captan ⁵	low (M 04)	4	0	0
Cevya	high (3)	4	3	5
Copper ⁵	low (M 01)	36	0	0
Dithane, Manzate, Penncozeb ⁵	low (M 03)	4	0	0
Excalia ⁵	high (7)	5	3	ND
Flint Extra ²	high (11) ³	5	5	5
Fontelis	high (7)	5	3	3
Gatten ⁵	high (U 13)	0	0	4
Inspire Super	medium (3/9)	5	5	5
Kenja	high (7)	5	3	4
Lime sulfur ^{5,7}	low (M 02)	3	5 ⁷	4 ⁸
Luna Sensation	medium (7/11)	5	3	5
Merivon	medium (7/11)	5	3	5
Omega, Lektivar ^{**,5}	medium (29)	4	2	NL
Ph-D, Oso	high (19)	3	0	4
Pristine	medium (7/11)	5	0	4
Procure ⁴	high (3)	5	5	5
Rally ⁵	high (3)	5	3	5
Regev	medium (3/BM 01)	5	5	5
Rhyme	high (3)	5	3	5
Rubigan**, Vintage ^{**,4}	high (3)	5	5	4
Scala	high (9) ³	4	4	2

Fungicide***	Resistance risk (FRAC group number) ¹	Scab		Powdery mildew
		protectant	eradicant	
Sovran	high (11) ³	4	4	4
Sulfur ⁶	low (M 02)	3	0	5
Syllit	medium (U12)	4	4	0
Tebucon, Toledo, etc.	high (3)	4	4	4
Tesaris, Sercadis**	high (7)	5	3	4
Topsin-M, T-Methyl, Incognito, Cercobin**	high (1) ³	4	4	4
Torino	high (U 6)	NL	NL	ND
Vangard	high (9) ³	4	4	4
Viathon ⁵	medium (3, P 07/33)	4	4	0
Ziram ⁵	low (M 03)	3	0	0

Rating: 5 = excellent and consistent, 4 = good and reliable, 3 = moderate and variable, 2 = limited and/or erratic, 1 = minimal and often ineffective, 0 = ineffective, NL = not on label, ND = no data

**	Not registered, label withdrawn or inactive in California.
***	Postharvest fruit registrations in California include: Academy, Alumni/TBZ, BioSpectra/CeraFruta, Penbotec/Pyrimethanil and Scholar/FDL.
1	Group numbers are assigned by the Fungicide Resistance Action Committee(http://www.frac.info/) (FRAC) according to different modes of action (for more information, see www.frac.info). NC = Not coded in FRAC. Fungicides with different group numbers are suitable to alternate in a resistance management program. In California, make no more than one application of a fungicide with a mode-of-action group number associated with high resistance risk before rotating to a fungicide with a different mode-of-action group number; for other fungicides, make no more than two consecutive applications before rotating to fungicide with a different mode-of-action group number.
2	High resistance potential to trifloxystrobin for apple and pear scab pathogen populations.
3	To reduce the risk of resistance development, start treatments with a fungicide with a multi-site mode of action; rotate or mix fungicides with different mode-of-action FRAC numbers for subsequent applications, use labeled rates (preferably the upper range), and limit the total number of applications per season. Cercobin is registered only on apples.
4	On pear, use only before white bud and after full bloom.
5	Labeled on apple only in California (not pear)
6	These materials show some efficacy and should be used in mixtures with antibiotics as a component of resistance management programs. Captan is registered on apples, whereas Dithane is registered on apples and pears.
	Though copper may be effective for scab and blight control under low disease pressure, copper products may cause fruit scarring or russetting.
8	"Burns out" scab twig lesions when applied at delayed dormant and disrupts pseudothecial (or ascostroma) development when applied to leaves in fall. CAUTION: Lime Sulfur is incompatible with most other pesticides when used after budbreak. Check before use. Sulfur products are also used to thin flowers because of their phytotoxicity and indirectly control fire blight.
9	In-season application eradicates powdery mildew.

Acknowledgment: Adaskaveg et al., 2025. [*Fungicides, Bactericides, Biocontrols, and Natural Products for Deciduous Tree Fruit and Nut, Citrus, Strawberry, and Vine Crops in California*\(/PDF/PMG/fungicideefficacytiming.pdf\)](#).



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